

BERLIK, Kazakhstan

WMO: 353760

Lat: 49.883N

Lon: 69.517E

Elev: 349

StdP: 97.20

Time Zone: 6.00 (E06)

Period: 94-08

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.8	-30.1	-35.5	0.1	-32.5	-32.9	0.2	-30.1	14.0	-8.0	11.9	-9.2	1.3	180	0.640

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	33.8	18.3	32.1	17.5	30.3	17.1	19.8	30.2	18.8	28.6	18.0	27.4	4.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.5	22.4	15.6	11.6	21.6	14.5	10.7	21.4	58.3	30.3	55.0	28.6	52.4	27.4	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	10.4	9.5	-35.0	37.1	1.7	1.8	-36.3	38.4	-37.3	39.4	-38.2	40.4	-39.4	41.7		
(5)			-34.9	22.0	1.8	3.1	-36.2	24.3	-37.2	26.1	-38.2	27.8	-39.4	30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.9	-16.1	-14.5	-7.6	6.6	14.7	19.7	21.4	20.1	13.3	5.6	-4.7	-12.9		
	DBStd	14.97	8.06	6.77	7.81	6.47	4.64	3.91	3.34	4.32	4.83	5.28	7.38	6.93		
	HDD10.0	3525	811	686	548	137	14	1	0	23	156	440	710			
	HDD18.3	5562	1069	919	806	353	128	29	11	32	162	396	690	968		
	CDD10.0	1291	0	0	1	34	160	291	352	313	122	18	0	0		
	CDD18.3	286	0	0	0	1	15	70	104	86	10	0	0	0		
	CDH23.3	3975	0	0	0	23	287	1044	1262	1137	212	10	0	0		
	CDH26.7	1564	0	0	0	4	76	427	502	486	69	1	0	0		
(14) Wind	WSAvg	3.6	3.4	4.4	3.8	4.0	3.8	3.7	3.5	3.2	3.1	3.2	3.5	3.7		
(15) Precipitation	PrecAvg	232	14	13	12	17	24	37	32	18	13	18	18	15		
	PrecMax	326	28	31	32	39	55	108	78	77	48	55	60	48		
	PrecMin	102	4	4	2	1	2	2	0	0	2	2	1	1		
	PrecStd	65	6	6	9	12	16	31	21	18	12	14	13	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.9	1.3	14.4	25.9	31.6	36.1	35.7	35.9	32.1	24.7	12.3	1.5		
		MCWB	0.3	0.4	8.6	14.5	16.7	18.0	20.3	18.4	15.9	12.0	7.7	0.5		
	2%	DB	-0.5	-0.2	8.5	22.6	28.8	33.1	33.2	33.6	28.4	20.9	9.0	-0.2		
		MCWB	-1.2	-0.9	5.2	12.2	15.4	17.6	18.5	18.3	14.5	10.6	6.1	-0.9		
	5%	DB	-2.6	-2.1	4.0	20.2	26.5	31.1	31.4	31.5	25.4	17.8	6.2	-1.6		
		MCWB	-3.4	-2.8	2.1	11.0	14.5	16.8	18.0	17.4	13.5	9.1	4.2	-2.4		
	10%	DB	-4.4	-4.2	1.7	17.3	24.3	29.1	29.5	29.3	22.9	14.6	4.0	-2.9		
		MCWB	-5.1	-4.8	0.5	9.5	13.6	16.3	17.5	16.7	12.5	7.7	2.2	-3.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.4	0.5	9.2	15.3	18.7	20.7	21.9	21.3	17.1	12.8	9.3	1.0		
		MCDB	0.7	1.1	13.4	24.3	28.2	31.7	31.5	31.4	30.8	21.4	11.4	1.4		
	2%	WB	-1.2	-0.7	5.4	13.1	16.6	18.9	20.1	19.2	15.4	11.4	6.2	-0.8		
		MCDB	-0.5	-0.1	8.1	21.5	26.2	30.4	30.7	30.5	25.7	19.3	8.3	-0.3		
	5%	WB	-3.3	-2.7	2.4	11.6	15.3	17.8	19.1	18.2	14.1	9.8	4.2	-2.4		
		MCDB	-2.7	-2.1	4.0	18.9	24.1	28.4	27.9	29.1	23.8	16.4	6.2	-1.7		
	10%	WB	-5.0	-4.8	0.6	10.0	14.2	16.9	18.4	17.3	12.9	8.3	2.3	-3.6		
		MCDB	-4.4	-4.2	1.7	16.5	22.7	26.6	26.5	27.2	21.9	14.0	3.9	-2.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	9.9	9.5	12.4	14.2	14.2	13.0	14.1	14.1	11.8	8.8	8.9		
		MCDBR	7.2	10.1	9.7	16.6	17.8	17.9	15.7	16.9	18.3	17.0	10.5	8.3		
		MCWBR	7.0	9.8	6.9	7.9	7.6	7.2	5.8	6.3	8.3	9.2	7.5	8.0		
		MCDBR	7.3	9.6	9.5	15.7	15.7	15.4	13.4	15.1	16.2	14.9	9.5	8.3		
	5% WB	MCWBR	7.2	9.4	7.0	8.1	7.7	6.9	5.6	6.7	8.0	8.7	7.2	8.1		
(40) Clear-Sky Solar Irradiance	taub		0.296	0.330	0.381	0.412	0.412	0.411	0.428	0.402	0.354	0.342	0.328	0.298		
	taud		2.075	2.002	2.021	2.127	2.195	2.263	2.224	2.282	2.366	2.317	2.188	2.098		
	Ebn at Noon		709	776	805	828	846	849	826	831	839	774	669	642		
	Edh at Noon		85	118	140	141	138	130	133	120	99	86	77	72		
(44) All-Sky Solar Radiation	RadAvg		1.54	2.71	3.98	5.02	6.32	6.66	6.20	5.59	4.24	2.40	1.18	1.04		
	RadStd		0.15	0.16	0.43	0.45	0.44	0.44	0.35	0.41	0.28	0.32	0.14	0.16		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page